

The Science of Seeing

Yellow Level: I'm Ready for a Challenge

Name: _____ Date: _____ Period: _____

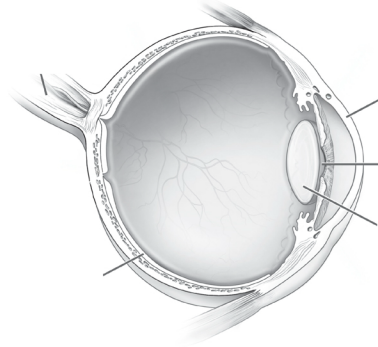
The Visual System

Vision isn't just about your eyes—it's a complex system involving light, specialized cells, nerve signals, and brain interpretation. Optical illusions reveal the limitations built into this system.



Activity 1: Eye Anatomy & Function

Label the eye diagram AND explain what each part does:



1. **Cornea:** _____
2. **Pupil:** _____
3. **Lens:** _____
4. **Retina:** _____
5. **Optic Nerve:** _____

Bonus: What's the difference between rods and cones?



Activity 2: Analyze the Illusions

For each illusion, identify: (1) What you see, (2) Why it happens, (3) Where in the visual system

Illusion A: Color Afterimage (Stare at colored image, then look at white paper)

What you see: _____

Why it happens: _____

Location (circle): Eye (retina) | Brain (visual cortex)

Illusion B: Müller-Lyer (Lines with arrows look different lengths)

What you see: _____

Why it happens: _____

Location (circle): Eye (retina) | Brain (visual cortex)

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The Blind Spot Experiment

Use the Blind Spot Test Card your teacher gave you. Close left eye, focus on +, move card closer until dot disappears.

Explain scientifically why this happens:

Compare Eye-Based vs. Brain-Based Illusions

Type	Where it happens	Example
Eye-based		
Brain-based		

Mini Museum Connection

The stereoscope exploits **binocular vision**—the fact that your two eyes see slightly different images. Your visual cortex combines these into depth perception.

Explain how this works in your own words:

Try the Perception Lab: minimuseumproject.com/exhibitions/seeing-is-deceiving/illusions

Share with Your Family!

Experiment: Find the blind spot in both eyes. Is it in the same place? (Hint: it's not!)

Discussion: Ask your family: "If your brain fills in your blind spot, what else might it be 'filling in' that isn't really there?"